

may exist.

Reader: What is this limit?

Author: It depends on the functions $f(x)$ and $g(x)$. Let us show, for example,

that $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$.

Note that the function $\frac{\sin x}{x}$ is not defined at $x=0$. This fact, however, does not interfere with searching for the limit of the function at $x=0$.

We shall start with the well-known inequalities:

$$\sin x < x < \tan x \quad (0 < x < \frac{\pi}{2}) \quad ??$$

An assumption that $0 < x < \frac{\pi}{2}$ will not affect the generality of our results. Dividing $\sin x$ by each term of these inequalities, we obtain.

$$1 > \frac{\sin x}{x} > \cos x \quad \text{hence} \quad 0 < (1 - \frac{\sin x}{x}) < (1 - \cos x)$$

$$\text{Next we take into account that } 1 - \cos x = 2 \sin^2 \frac{x}{2} < 2 \sin \frac{x}{2} < 2 \frac{x}{2} = x \quad ??$$

$$\text{Thus we have } 0 < (1 - \frac{\sin x}{x}) < x \quad \text{or, } -x < (1 - \frac{\sin x}{x}) < 0$$

whence $|1 - \frac{\sin x}{x}| < |x|$. We thus arrive at the following inequality

$$\text{valid for } |x| < \frac{\pi}{2}: \quad \left| \frac{\sin x}{x} - 1 \right| < |x| \quad (1)$$

By using this inequality, we can easily prove that the function $\frac{\sin x}{x}$ has the limit at $x=0$, and this limit is unity. It will be convenient to use definition 1 for the limit of function at a point.

Select an arbitrary $\varepsilon > 0$, demanding for the sake of simplicity that $\varepsilon < \frac{\pi}{2}$. For δ , it is sufficient to take $\delta = \varepsilon$ since, according to (1), the condition $|x-0| < \delta$ immediately leads to

$$\left| \frac{\sin x}{x} - 1 \right| < \delta = \varepsilon$$

Thus, unity is indeed the limit of the function $\frac{\sin x}{x}$ at $x=0$.